

99.500 Applied Mathematics for Engineering

Homework 1

February 4, 2026

1. A semi-infinite slab is initially at a uniform temperature T_i . At time $t = 0$, the temperature at the surface begins to increase linearly with time, following the relation:

$$T(x = 0, t) = T_i + At$$

where A is a constant. Within the slab, heat transfer occurs solely by conduction, governed by Fourier's law.

- (a) Using the shell balance technique, derive the governing equation for the non-steady-state temperature distribution $T(x, t)$.
 - (b) Determine the appropriate boundary and initial conditions for this problem.
2. A water stream with a constant volumetric flow rate Q is fed into a pipe of radius R with a permeable wall, as illustrated below. The pipe is embedded in a porous medium that extends infinitely in the radial direction but has a finite height L . The water contains a dissolved reagent at an initial concentration C_F which undergoes a first-order reaction on the surface of the solid particles within the porous medium. The reaction rate is given by $r_s = kC \left(\frac{\text{mole}}{\text{cm}^2\text{s}} \right)$. The molar flux of the reagent N is expressed as

$$N = Cv - D \frac{dC}{dr}$$

where v is the superficial velocity and D is the effective diffusivity.

- (a) Using the shell balance technique, show that the steady-state concentration distribution is governed by the equation:

$$r^2 \frac{d^2 C}{dr^2} + r \left(1 - \frac{rv}{D} \right) \frac{dC}{dr} - \frac{ak}{D} r^2 C = 0$$

where a is the specific interfacial area of the porous solid $\left(\frac{\text{cm}^2}{\text{cm}^3} \right)$. Clearly state any assumptions made in the derivation. [Hint: For dilute solutions, mass continuity requires $\frac{d}{dr}(rv) = 0$].

- (b) Identify the appropriate boundary conditions for the governing equation. Hint: Consider the two physical boundaries in the system.
- (c) Nondimensionalize the governing equation and its boundary conditions.

